Part A: Classwork

79 questions solved in the live

Every question here is an actual CAT question from 2017 to 2025, in the order we solved them in the Arithmetic 1 live. Try each one yourself first, then watch its solution in the video. The answer key for the full handout is at the end.

Percentages (17 questions)

1. The income of Amala is 20% more than that of Bimala and 20% less than that of Kamala. If Kamala's income goes down by 4% and Bimala's goes up by 10%, then the percentage by which Kamala's income would exceed Bimala's is nearest to *(CAT 2019, Slot 1)*

A. 31 B. 29 C. 28 D. 32

2. Out of the shirts produced in a factory, 15% are defective, while 20% of the rest are sold in the domestic market. If the remaining 8840 shirts are left for export, then the number of shirts produced in the factory is *(CAT 2017, Slot 2)*

A. 13600 B. 13000 C. 13400 D. 14000

3. The number of girls appearing for an admission test is twice the number of boys. If 30% of the girls and 45% of the boys get admission, the percentage of candidates who do not get admission is *(CAT 2017, Slot 1)*

A. 35 B. 50 C. 60 D. 65

4. In an examination, the score of A was 10% less than that of B, the score of B was 25% more than that of C, and the score of C was 20% less than that of D. If A scored 72, then the score of D was *(CAT 2019, Slot 2)*

5. In a village, the production of food grains increased by 40% and the per capita production of food grains increased by 27% during a certain period. The percentage by which the population of the village increased during the same period is nearest to *(CAT 2017, Slot 2)*

A. 16 B. 13 C. 10 D. 7

6. In a class, 60% of the students are girls and the rest are boys. There are 30 more girls than boys. If 68% of the students, including 30 boys, pass an examination, the percentage of the girls who do not pass is *(CAT 2019, Slot 1)*

7. In May, John bought the same amount of rice and the same amount of wheat as he had bought in April, but spent ₹150 more due to price increase of rice and wheat by 20% and 12%, respectively. If John had spent ₹450 on rice in April, then how much did he spend on wheat in May? *(CAT 2020, Slot 2)*

A. 590 B. 580 C. 560 D. 570

8. The total of male and female populations in a city increased by 25% from 1970 to 1980. During the same period, the male population increased by 40% while the female population increased by 20%. From 1980 to 1990, the female population increased by 25%. In 1990, if the female population is twice the male population, then the percentage increase in the total of male and female populations in the city from 1970 to 1990 is *(CAT 2021, Slot 3)*

A. 68.75 B. 68.50 C. 68.25 D. 69.25

9. After two successive increments, Gopal's salary became 187.5% of his initial salary. If the percentage of salary increase in the second increment was twice of that in the first increment, then the percentage of salary increase in the first increment was *(CAT 2024, Slot 3)*

A. 27.5 B. 30 C. 25 D. 20

10. Raj invested ₹10000 in a fund. At the end of first year, he incurred a loss but his balance was more than ₹5000. This balance, when invested for another year, grew and the percentage of growth in the second year was five times the percentage of loss in the first year. If the gain of Raj from the initial investment over the two year period is 35%, then the percentage of loss in the first year is *(CAT 2021, Slot 2)*

A. 15 B. 5 C. 10 D. 70

11. The salaries of three friends Sita, Gita and Mita are initially in the ratio 5 : 6 : 7, respectively. In the first year, they get salary hikes of 20%, 25% and 20%, respectively. In the second year, Sita and Mita get salary hikes of 40% and 25%, respectively, and the salary of Gita becomes equal to the mean salary of the three friends. The salary hike of Gita in the second year is *(CAT 2023, Slot 1)*

A. 25% B. 28% C. 26% D. 30%

12. In September, the incomes of Kamal, Amal and Vimal are in the ratio 8 : 6 : 5. They rent a house together, and Kamal pays 15%, Amal pays 12% and Vimal pays 18% of their respective incomes to cover the total house rent in that month. In October, the house rent remains unchanged while their incomes increase by 10%, 12% and 15%, respectively. In October, the percentage of their total income that will be paid as house rent, is nearest to *(CAT 2024, Slot 1)*

A. 14.84 B. 12.75 C. 13.26 D. 15.18

13. In a tournament, a team has played 40 matches so far and won 30% of them. If they win 60% of the remaining matches, their overall win percentage will be 50%. Suppose they win 90% of the remaining matches, then the total number of matches won by the team in the tournament will be *(CAT 2021, Slot 3)*

A. 86 B. 84 C. 78 D. 80

14. The population of a town in 2020 was 100000. The population decreased by y% from the year 2020 to 2021, and increased by x% from the year 2021 to 2022, where x and y are two natural numbers. If population in 2022 was greater than the population in 2020 and the difference between x and y is 10, then the lowest possible population of the town in 2021 was *(CAT 2023, Slot 3)*

A. 74000 B. 75000 C. 72000 D. 73000

15. In a class, there were more than 10 boys and a certain number of girls. After 40% of the girls and 60% of the boys left the class, the remaining number of girls was 8 more than the remaining number of boys. Then, the minimum possible number of students initially in the class was *(CAT 2025, Slot 1)*

16. A fruit seller has a stock of mangoes, bananas and apples with at least one fruit of each type. At the beginning of a day, the number of mangoes make up 40% of his stock. That day, he sells half of the mangoes, 96 bananas and 40% of the apples. At the end of the day, he ends up selling 50% of the fruits. The smallest possible total number of fruits in the stock at the beginning of the day is *(CAT 2023, Slot 3)*

17. In a group of 250 students, the percentage of girls was at least 44% and at most 60%. The rest of the students were boys. Each student opted for either swimming or running or both. If 50% of the boys and 80% of the girls opted for swimming while 70% of the boys and 60% of the girls opted for running, then the minimum and maximum possible number of students who opted for both swimming and running, are *(CAT 2024, Slot 3)*

A. 72 and 80, respectively
B. 75 and 90, respectively
C. 72 and 88, respectively
D. 75 and 96, respectively

Profit & Loss (17 questions)

18. If a seller gives a discount of 15% on retail price, she still makes a profit of 2%. Which of the following ensures that she makes a profit of 20%? *(CAT 2017, Slot 1)*

A. Give a discount of 5% on retail price
B. Give a discount of 2% on retail price
C. Increase the retail price by 2%
D. Sell at retail price

19. Gopi marks a price on a product in order to make 20% profit. Ravi gets 10% discount on this marked price, and thus saves Rs 15. Then, the profit, in rupees, made by Gopi by selling the product to Ravi, is *(CAT 2024, Slot 3)*

A. 25 B. 10 C. 15 D. 20

20. The selling price of a product is fixed to ensure 40% profit. If the product had cost 40% less and had been sold for 5 rupees less, then the resulting profit would have been 50%. The original selling price, in rupees, of the product is *(CAT 2024, Slot 1)*

A. 10 B. 15 C. 20 D. 14

21. The manufacturer of a table sells it to a wholesale dealer at a profit of 10%. The wholesale dealer sells the table to a retailer at a profit of 30%. Finally, the retailer sells it to a customer at a profit of 50%. If the customer pays Rs 4290 for the table, then its manufacturing cost (in Rs) is *(CAT 2017, Slot 2)*

A. 1500 B. 2000 C. 2500 D. 3000

22. Minu purchases a pair of sunglasses at Rs.1000 and sells to Kanu at 20% profit. Then, Kanu sells it back to Minu at 20% loss. Finally, Minu sells the same pair of sunglasses to Tanu. If the total profit made by Minu from all her transactions is Rs.500, then the percentage of profit made by Minu when she sold the pair of sunglasses to Tanu is *(CAT 2023, Slot 2)*

A. 35.42% B. 31.25% C. 52% D. 26%

23. Bina incurs 19% loss when she sells a product at Rs. 4860 to Shyam, who in turn sells this product to Hari. If Bina would have sold this product to Shyam at the purchase price of Hari, she would have obtained 17% profit. Then, the profit, in rupees, made by Shyam is *(CAT 2024, Slot 2)*

24. On selling a pen at 5% loss and a book at 15% gain, Karim gains Rs. 7. If he sells the pen at 5% gain and the book at 10% gain, he gains Rs. 13. What is the cost price of the book in Rupees? *(CAT 2019, Slot 1)*

A. 80 B. 85 C. 100 D. 95

25. Mukesh purchased 10 bicycles in 2017, all at the same price. He sold six of these at a profit of 25% and the remaining four at a loss of 25%. If he made a total profit of Rs. 2000, then his purchase price of a bicycle, in Rupees, was *(CAT 2019, Slot 2)*

A. 2000 B. 6000 C. 8000 D. 4000

26. A person spent Rs 50000 to purchase a desktop computer and a laptop computer. He sold the desktop at 20% profit and the laptop at 10% loss. If overall he made a 2% profit then the purchase price, in rupees, of the desktop is *(CAT 2020, Slot 1)*

27. Gita sells two objects A and B at the same price such that she makes a profit of 20% on object A and a loss of 10% on object B. If she increases the selling price such that objects A and B are still sold at an equal price and a profit of 10% is made on object B, then the profit made on object A will be nearest to *(CAT 2023, Slot 1)*

A. 42% B. 47% C. 45% D. 49%

28. Anil buys 12 toys and labels each with the same selling price. He sells 8 toys initially at 20% discount on the labeled price. Then he sells the remaining 4 toys at an additional 25% discount on the discounted price. Thus, he gets a total of Rs 2112, and makes a 10% profit. With no discounts, his percentage of profit would have been *(CAT 2020, Slot 2)*

A. 60 B. 55 C. 50 D. 54

29. If Fatima sells 60 identical toys at a 40% discount on the printed price, then she makes 20% profit. Ten of these toys are destroyed in fire. While selling the rest, how much discount should be given on the printed price so that she can make the same amount of profit? *(CAT 2017, Slot 1)*

A. 30% B. 25% C. 24% D. 28%

30. A shopkeeper offers a discount of 22% on the marked price of each chair, and gives 13 chairs to a customer for the discounted price of 12 chairs to earn a profit of 26% on the transaction. If the cost price of each chair is Rs 100, then the marked price, in rupees, of each chair is *(CAT 2025, Slot 1)*

31. A merchant purchases a cloth at a rate of Rs.100 per meter and receives 5 cm length of cloth free for every 100 cm length of cloth purchased by him. He sells the same cloth at a rate of Rs.110 per meter but cheats his customers by giving 95 cm length of cloth for every 100 cm length of cloth purchased by the customers. If the merchant provides a 5% discount, the resulting profit earned by him is *(CAT 2023, Slot 3)*

A. 4.2% B. 15.5% C. 16% D. 9.7%

32. An item with a cost price of Rs. 1650 is sold at a certain discount on a fixed marked price to earn a profit of 20% on the cost price. If the discount was doubled, the profit would have been Rs. 110. The rate of discount, in percentage, at which the profit percentage would be equal to the rate of discount, is nearest to *(CAT 2025, Slot 2)*

A. 12 B. 14 C. 16 D. 18

33. Amal purchases some pens at ₹8 each. To sell these, he hires an employee at a fixed wage. He sells 100 of these pens at ₹12 each. If the remaining pens are sold at ₹11 each, then he makes a net profit of ₹300, while he makes a net loss of ₹300 if the remaining pens are sold at ₹9 each. The wage of the employee, in INR, is *(CAT 2021, Slot 1)*

34. Jayant bought a certain number of white shirts at the rate of Rs 1000 per piece and a certain number of blue shirts at the rate of Rs 1125 per piece. For each shirt, he then set a fixed market price which was 25% higher than the average cost of all the shirts. He sold all the shirts at a discount of 10% and made a total profit of Rs 51000. If he bought both colors of shirts, then the maximum possible total number of shirts that he could have bought is *(CAT 2023, Slot 2)*

Simple & Compound Interest (17 questions)

35. At a certain simple rate of interest, a given sum amounts to Rs 13920 in 3 years, and to Rs 18960 in 6 years and 6 months. If the same given sum had been invested for 2 years at the same rate as before but with interest compounded every 6 months, then the total interest earned, in rupees, would have been nearest to *(CAT 2025, Slot 1)*

A. 3096 B. 3150 C. 3180 D. 3221

36. Veeru invested Rs 10000 at 5% simple annual interest, and exactly after two years, Joy invested Rs 8000 at 10% simple annual interest. How many years after Veeru's investment, will their balances, i.e., principal plus accumulated interest, be equal? *(CAT 2020, Slot 1)*

37. Alex invested his savings in two parts. The simple interest earned on the first part at 15% per annum for 4 years is the same as the simple interest earned on the second part at 12% per annum for 3 years. Then, the percentage of his savings invested in the first part is *(CAT 2022, Slot 1)*

A. 62.5% B. 37.5% C. 60% D. 40%

38. Mr. Pinto invests one-fifth of his capital at 6%, one-third at 10% and the remaining at 1%, each rate being simple interest per annum. Then, the minimum number of years required for the cumulative interest income from these investments to equal or exceed his initial capital is *(CAT 2022, Slot 2)*

39. A person invested a total amount of Rs 15 lakh. A part of it was invested in a fixed deposit earning 6% annual interest, and the remaining amount was invested in two other deposits in the ratio 2 : 1, earning annual interest at the rates of 4% and 3%, respectively. If the total annual interest income is Rs 76000 then the amount (in Rs lakh) invested in the fixed deposit was *(CAT 2019, Slot 1)*

40. An amount of Rs 10000 is deposited in bank A for a certain number of years at a simple interest of 5% per annum. On maturity, the total amount received is deposited in bank B for another 5 years at a simple interest of 6% per annum. If the interests received from bank A and bank B are in the ratio 10 : 13, then the investment period, in years, in bank A is *(CAT 2024, Slot 1)*

A. 4 B. 5 C. 3 D. 6

41. Gopal borrows Rs. X from Ankit at 8% annual interest. He then adds Rs. Y of his own money and lends Rs. X+Y to Ishan at 10% annual interest. At the end of the year, after returning Ankit's dues, the net interest retained by Gopal is the same as that accrued to Ankit. On the other hand, had Gopal lent Rs. X+2Y to Ishan at 10%, then the net interest retained by him would have increased by Rs. 150. If all interests are compounded annually, then find the value of X + Y. *(CAT 2018, Slot 2)*

42. A person invested a certain amount of money at 10% annual interest, compounded half-yearly. After one and a half years, the interest and principal together became Rs 18522. The amount, in rupees, that the person had invested is *(CAT 2020, Slot 3)*

43. For the same principal amount, the compound interest for two years at 5% per annum exceeds the simple interest for three years at 3% per annum by Rs 1125. Then the principal amount in rupees is *(CAT 2020, Slot 2)*

44. Amal invests Rs 12000 at 8% interest, compounded annually, and Rs 10000 at 6% interest, compounded semi-annually, both investments being for one year. Bimal invests his money at 7.5% simple interest for one year. If Amal and Bimal get the same amount of interest, then the amount, in Rupees, invested by Bimal is *(CAT 2019, Slot 2)*

45. Aman invests Rs 4000 in a bank at a certain rate of interest, compounded annually. If the ratio of the value of the investment after 3 years to the value of the investment after 5 years is 25 : 36, then the minimum number of years required for the value of the investment to exceed Rs 20000 is *(CAT 2024, Slot 3)*

46. Anil invests some money at a fixed rate of interest, compounded annually. If the interests accrued during the second and third year are ₹806.25 and ₹866.72, respectively, the interest accrued, in INR, during the fourth year is nearest to *(CAT 2021, Slot 1)*

A. 931.72 B. 926.84 C. 929.48 D. 934.65

47. Bank A offers 6% interest rate per annum compounded half yearly. Bank B and Bank C offer simple interest but the annual interest rate offered by Bank C is twice that of Bank B. Raju invests a certain amount in Bank B for a certain period and Rupa invests ₹10,000 in Bank C for twice that period. The interest that would accrue to Raju during that period is equal to the interest that would have accrued had he invested the same amount in Bank A for one year. The interest accrued, in INR, to Rupa is *(CAT 2021, Slot 3)*

A. 2436 B. 3436 C. 2346 D. 1436

48. John borrowed Rs. 2,10,000 from a bank at an interest rate of 10% per annum, compounded annually. The loan was repaid in two equal instalments, the first after one year and the second after another year. The first instalment was interest of one year plus part of the principal amount, while the second was the rest of the principal amount plus due interest thereon. Then each instalment, in Rs., is *(CAT 2018, Slot 1)*

49. A loan of Rs 1000 is fully repaid by two installments of Rs 530 and Rs 594, paid at the end of first and second year, respectively. If the interest is compounded annually, then the rate of interest, in percentage, is *(CAT 2025, Slot 2)*

A. 8 B. 10 C. 11 D. 9

50. Anil borrows Rs 2 lakhs at an interest rate of 8% per annum, compounded half-yearly. He repays Rs 10320 at the end of the first year and closes the loan by paying the outstanding amount at the end of the third year. Then, the total interest, in rupees, paid over the three years is nearest to *(CAT 2023, Slot 2)*

A. 45311 B. 51311 C. 33130 D. 40991

51. Anil invests Rs 22000 for 6 years in a scheme with 4% interest per annum, compounded half-yearly. Separately, Sunil invests a certain amount in the same scheme for 5 years, and then reinvests the entire amount he receives at the end of 5 years, for one year at 10% simple interest. If the amounts received by both at the end of 6 years are equal, then the initial investment, in rupees, made by Sunil is *(CAT 2024, Slot 2)*

A. 20860 B. 20640 C. 20480 D. 20808

Ratio, Proportion & Variation (17 questions)

52. The ratio of the number of coins in boxes A and B was 17 : 7. After 108 coins were shifted from box A to box B, this ratio became 37 : 20. The number of coins that needs to be shifted further from A to B, to make this ratio 1 : 1, is *(CAT 2025, Slot 3)*

53. If a, b, c are three positive integers such that a and b are in the ratio 3 : 4 while b and c are in the ratio 2 : 1, then which one of the following is a possible value of (a + b + c)? *(CAT 2017, Slot 2)*

A. 201 B. 205 C. 207 D. 210

54. A sum of money is split among Amal, Sunil and Mita so that the ratio of the shares of Amal and Sunil is 3 : 2, while the ratio of the shares of Sunil and Mita is 4 : 5. If the difference between the largest and the smallest of these three shares is Rs 400, then Sunil's share, in rupees, is *(CAT 2020, Slot 2)*

55. Suppose, C1, C2, C3, C4, and C5 are five companies. The profits made by C1, C2, and C3 are in the ratio 9 : 10 : 8 while the profits made by C2, C4, and C5 are in the ratio 18 : 19 : 20. If C5 has made a profit of Rs 19 crore more than C1, then the total profit (in Rs) made by all five companies is *(CAT 2017, Slot 1)*

A. 438 crore B. 435 crore C. 348 crore D. 345 crore

56. The scores of Amal and Bimal in an examination are in the ratio 11 : 14. After an appeal, their scores increase by the same amount and their new scores are in the ratio 47 : 56. The ratio of Bimal's new score to that of his original score is *(CAT 2018, Slot 2)*

A. 3 : 2 B. 4 : 3 C. 5 : 4 D. 8 : 5

57. Raju and Lalitha originally had marbles in the ratio 4 : 9. Then Lalitha gave some of her marbles to Raju. As a result, the ratio of the number of marbles with Raju to that with Lalitha became 5 : 6. What fraction of her original number of marbles was given by Lalitha to Raju? *(CAT 2018, Slot 1)*

A. 1/4 B. 1/5 C. 6/19 D. 7/33

58. The ratio of the number of students in the morning shift and afternoon shift of a school was 13 : 9. After 21 students moved from the morning shift to the afternoon shift, this ratio became 19 : 14. Next, some new students joined the morning and afternoon shifts in the ratio 3 : 8 and then the ratio of the number of students in the morning shift and the afternoon shift became 5 : 4. The number of new students who joined is *(CAT 2025, Slot 1)*

A. 110 B. 88 C. 121 D. 99

59. A fruit seller has a total of 187 fruits consisting of apples, mangoes and oranges. The number of apples and mangoes are in the ratio 5 : 2. After she sells 75 apples, 26 mangoes and half of the oranges, the ratio of number of unsold apples to number of unsold oranges becomes 3 : 2. The total number of unsold fruits is *(CAT 2024, Slot 1)*

60. When Rajesh's age was same as the present age of Garima, the ratio of their ages was 3 : 2. When Garima's age becomes the same as the present age of Rajesh, the ratio of the ages of Rajesh and Garima will become *(CAT 2024, Slot 2)*

A. 4 : 3 B. 2 : 1 C. 3 : 2 D. 5 : 4

61. The salaries of Ramesh, Ganesh and Rajesh were in the ratio 6:5:7 in 2010, and in the ratio 3:4:3 in 2015. If Ramesh's salary increased by 25% during 2010-2015, then the percentage increase in Rajesh's salary during this period is closest to *(CAT 2019, Slot 2)*

A. 8 B. 10 C. 7 D. 9

62. In a village, the ratio of number of males to females is 5 : 4. The ratio of number of literate males to literate females is 2 : 3. The ratio of the number of illiterate males to illiterate females is 4 : 3. If 3600 males in the village are literate, then the total number of females in the village is *(CAT 2022, Slot 1)*

63. Rajesh and Vimal own 20 hectares and 30 hectares of agricultural land, respectively, which are entirely covered by wheat and mustard crops. The cultivation area of wheat and mustard in the land owned by Vimal are in the ratio of 5 : 3. If the total cultivation area of wheat and mustard are in the ratio 11 : 9, then the ratio of cultivation area of wheat and mustard in the land owned by Rajesh is *(CAT 2024, Slot 3)*

A. 1 : 1 B. 7 : 9 C. 4 : 3 D. 3 : 7

64. The ratio of expenditures of Lakshmi and Meenakshi is 2 : 3, and the ratio of income of Lakshmi to expenditure of Meenakshi is 6 : 7. If excess of income over expenditure is saved by Lakshmi and Meenakshi, and the ratio of their savings is 4 : 9, then the ratio of their incomes is *(CAT 2025, Slot 2)*

A. 5 : 6 B. 3 : 5 C. 7 : 8 D. 2 : 1

65. The amount Neeta and Geeta together earn in a day equals what Sita alone earns in 6 days. The amount Sita and Neeta together earn in a day equals what Geeta alone earns in 2 days. The ratio of the daily earnings of the one who earns the most to that of the one who earns the least is *(CAT 2021, Slot 1)*

A. 7 : 3 B. 11 : 3 C. 11 : 7 D. 3 : 2

66. Anil, Bobby and Chintu jointly invest in a business and agree to share the overall profit in proportion to their investments. Anil's share of investment is 70%. His share of profit decreases by ₹420 if the overall profit goes down from 18% to 15%. Chintu's share of profit increases by ₹80 if the overall profit goes up from 15% to 17%. The amount, in INR, invested by Bobby is *(CAT 2021, Slot 2)*

A. 2400 B. 2200 C. 2000 D. 1800

67. One part of a hostel's monthly expenses is fixed, and the other part is proportional to the number of its boarders. The hostel collects ₹1600 per month from each boarder. When the number of boarders is 50, the profit of the hostel is ₹200 per boarder, and when the number of boarders is 75, the profit of the hostel is ₹250 per boarder. When the number of boarders is 80, the total profit of the hostel, in INR, will be *(CAT 2021, Slot 3)*

A. 20800 B. 20200 C. 20500 D. 20000

68. The price of a precious stone is directly proportional to the square of its weight. Sita has a precious stone weighing 18 units. If she breaks it into four pieces with each piece having distinct integer weight, then the difference between the highest and lowest possible values of the total price of the four pieces will be 288000. Then, the price of the original precious stone is *(CAT 2023, Slot 2)*

A. 1296000 B. 1944000 C. 972000 D. 1620000

Averages (11 questions)

69. In a company, 20% of the employees work in the manufacturing department. If the total salary obtained by all the manufacturing employees is one-sixth of the total salary obtained by all the employees in the company, then the ratio of the average salary obtained by the manufacturing employees to the average salary obtained by the non-manufacturing employees is *(CAT 2023, Slot 2)*

A. 4 : 5 B. 5 : 6 C. 5 : 4 D. 6 : 5

70. Dick is thrice as old as Tom and Harry is twice as old as Dick. If Dick's age is 1 year less than the average age of all three, then Harry's age, in years, is *(CAT 2020, Slot 3)*

71. The average number of copies of a book sold per day by a shopkeeper is 60 in the initial seven days and 63 in the initial eight days, after the book launch. On the ninth day, she sells 11 copies less than the eighth day, and the average number of copies sold per day from second day to ninth day becomes 66. The number of copies sold on the first day of the book launch is *(CAT 2025, Slot 2)*

72. Ramesh and Gautam are among 22 students who write an examination. Ramesh scores 82.5. The average score of the 21 students other than Gautam is 62. The average score of all the 22 students is one more than the average score of the 21 students other than Ramesh. The score of Gautam is *(CAT 2019, Slot 1)*

A. 53 B. 51 C. 48 D. 49

73. There are three persons A, B and C in a room. If a person D joins the room, the average weight of the persons in the room reduces by x kg. Instead of D, if person E joins the room, the average weight of the persons in the room increases by 2x kg. If the weight of E is 12 kg more than that of D, then the value of x is *(CAT 2023, Slot 3)*

A. 1.5 B. 2 C. 0.5 D. 1

74. The average weight of students in a class increases by 600 gm when some new students join the class. If the average weight of the new students is 3 kg more than the average weight of the original students, then the ratio of the number of original students to the number of new students is *(CAT 2022, Slot 1)*

A. 1 : 2 B. 3 : 1 C. 1 : 4 D. 4 : 1

75. A CAT aspirant appears for a certain number of tests. His average score increases by 1 if the first 10 tests are not considered, and decreases by 1 if the last 10 tests are not considered. If his average scores for the first 10 and the last 10 tests are 20 and 30, respectively, then the total number of tests taken by him is *(CAT 2018, Slot 1)*

76. A company has 40 employees whose names are listed in a certain order. In the year 2022, the average bonus of the first 30 employees was Rs. 40000, of the last 30 employees was Rs. 60000, and of the first 10 and last 10 employees together was Rs. 50000. Next year, the average bonus of the first 10 employees increased by 100%, of the last 10 employees increased by 200% and of the remaining employees was unchanged. Then, the average bonus, in rupees, of all the 40 employees together in the year 2023 was *(CAT 2024, Slot 2)*

A. 90000 B. 95000 C. 80000 D. 85000

77. The average salary of 5 managers and 25 engineers in a company is 60000 rupees. If each of the managers received 20% salary increase while the salary of the engineers remained unchanged, the average salary of all 30 employees would have increased by 5%. The average salary, in rupees, of the engineers is *(CAT 2025, Slot 3)*

A. 54000 B. 40000 C. 45000 D. 50000

78. The average of 30 integers is 5. Among these 30 integers, there are exactly 20 which do not exceed 5. What is the highest possible value of the average of these 20 integers? *(CAT 2019, Slot 2)*

A. 3.5 B. 5 C. 4.5 D. 4

79. A class consists of 20 boys and 30 girls. In the mid-semester examination, the average score of the girls was 5 higher than that of the boys. In the final exam, however, the average score of the girls dropped by 3 while the average score of the entire class increased by 2. The increase in the average score of the boys is *(CAT 2017, Slot 1)*

A. 9.5 B. 10 C. 4.5 D. 6

Part B: Practice

44 more CAT questions on the same topics

These are the remaining CAT questions on the same topics that we did not solve in the live. Solve them on paper at about 3 minutes a question. Simple & Compound Interest has no extra questions because we solved all of them in class.

Percentages (12 questions)

80. Arun's present age in years is 40% of Barun's. In another few years, Arun's age will be half of Barun's. By what percentage will Barun's age increase during this period? *(CAT 2017, Slot 1)*

81. Ravi invests 50% of his monthly savings in fixed deposits. Thirty percent of the rest of his savings is invested in stocks and the rest goes into Ravi's savings bank account. If the total amount deposited by him in the bank (for savings account and fixed deposits) is Rs 59500, then Ravi's total monthly savings (in Rs) is *(CAT 2017, Slot 1)*

82. In an examination, the maximum possible score is N while the pass mark is 45% of N. A candidate obtains 36 marks, but falls short of the pass mark by 68%. Which one of the following is then correct? *(CAT 2018, Slot 1)*

A. $N \leq 200$ B. $243 \leq N \leq 252$ C. $N \geq 253$ D. $201 \leq N \leq 242$

83. Meena scores 40% in an examination and after review, even though her score is increased by 50%, she fails by 35 marks. If her post-review score is increased by 20%, she will have 7 marks more than the passing score. The percentage score needed for passing the examination is *(CAT 2019, Slot 1)*

A. 75 B. 80 C. 60 D. 70

84. In 2010, a library contained a total of 11500 books in two categories - fiction and non-fiction. In 2015, the library contained a total of 12760 books in these two categories. During this period, there was 10% increase in the fiction category while there was 12% increase in the non-fiction category. How many fiction books were in the library in 2015? *(CAT 2019, Slot 2)*

A. 6600 B. 6160 C. 6000 D. 5500

85. In a group of people, 28% of the members are young while the rest are old. If 65% of the members are literates, and 25% of the literates are young, then the percentage of old people among the illiterates is nearest to *(CAT 2020, Slot 1)*

A. 59 B. 62 C. 66 D. 55

86. In the final examination, Bishnu scored 52% and Asha scored 64%. The marks obtained by Bishnu is 23 less, and that by Asha is 34 more than the marks obtained by Ramesh. The marks obtained by Geeta, who scored 84%, is *(CAT 2020, Slot 3)*

A. 439 B. 399 C. 357 D. 417

87. Identical chocolate pieces are sold in boxes of two sizes, small and large. The large box is sold for twice the price of the small box. If the selling price per gram of chocolate in the large box is 12% less than that in the small box, then the percentage by which the weight of chocolate in the large box exceeds that in the small box is nearest to *(CAT 2021, Slot 1)*

A. 127 B. 135 C. 124 D. 144

88. A box has 450 balls, each either white or black, there being as many metallic white balls as metallic black balls. If 40% of the white balls and 50% of the black balls are metallic, then the number of non-metallic balls in the box is *(CAT 2021, Slot 2)*

89. In an election, there were four candidates and 80% of the registered voters casted their votes. One of the candidates received 30% of the casted votes while the other three candidates received the remaining casted votes in the proportion 1 : 2 : 3. If the winner of the election received 2512 votes more than the candidate with the second highest votes, then the number of registered voters was *(CAT 2022, Slot 2)*

A. 40192 B. 60288 C. 50240 D. 62800

90. Kamala divided her investment of Rs 100000 between stocks, bonds, and gold. Her investment in bonds was 25% of her investment in gold. With annual returns of 10%, 6%, 8% on stocks, bonds, and gold, respectively, she gained a total amount of Rs 8200 in one year. The amount, in rupees, that she gained from the bonds, was *(CAT 2025, Slot 1)*

91. A certain amount of money was divided among Pinu, Meena, Rinu and Seema. Pinu received 20% of the total amount and Meena received 40% of the remaining amount. If Seema received 20% less than Pinu, the ratio of the amounts received by Pinu and Rinu is *(CAT 2025, Slot 2)*

A. 8 : 5 B. 5 : 8 C. 2 : 1 D. 1 : 2

Profit & Loss (7 questions)

92. In a market, the price of medium quality mangoes is half that of good mangoes. A shopkeeper buys 80 kg good mangoes and 40 kg medium quality mangoes from the market and then sells all these at a common price which is 10% less than the price at which he bought the good ones. His overall profit is *(CAT 2017, Slot 1)*

A. 6% B. 8% C. 10% D. 12%

93. Mayank buys some candies for Rs 15 a dozen and an equal number of different candies for Rs 12 a dozen. He sells all for Rs 16.50 a dozen and makes a profit of Rs 150. How many dozens of candies did he buy altogether? *(CAT 2017, Slot 2)*

A. 50 B. 30 C. 25 D. 45

94. A shopkeeper sells two tables, each procured at cost price p, to Amal and Asim at a profit of 20% and at a loss of 20%, respectively. Amal sells his table to Bimal at a profit of 30%, while Asim sells his table to Barun at a loss of 30%. If the amounts paid by Bimal and Barun are x and y, respectively, then $(x - y)/p$ equals *(CAT 2019, Slot 2)*

A. 1 B. 1.2 C. 0.7 D. 0.50

95. A man buys 35 kg of sugar and sets a marked price in order to make a 20% profit. He sells 5 kg at this price, and 15 kg at a 10% discount. Accidentally, 3 kg of sugar is wasted. He sells the remaining sugar by raising the marked price by p percent so as to make an overall profit of 15%. Then p is nearest to *(CAT 2020, Slot 3)*

A. 35 B. 31 C. 22 D. 25

96. Amal buys 110 kg of syrup and 120 kg of juice, syrup being 20% less costly than juice, per kg. He sells 10 kg of syrup at 10% profit and 20 kg of juice at 20% profit. Mixing the remaining juice and syrup, Amal sells the mixture at ₹308.32 per kg and makes an overall profit of 64%. Then, Amal's cost price for syrup, in rupees per kg, is *(CAT 2022, Slot 1)*

97. Ankita buys 4 kg cashews, 14 kg peanuts and 6 kg almonds when the cost of 7 kg cashews is the same as that of 30 kg peanuts or 9 kg almonds. She mixes all the three nuts and marks a price for the mixture in order to make a profit of ₹1752. She sells 4 kg of the mixture at this marked price and the remaining at a 20% discount on the marked price, thus making a total profit of ₹744. Then the amount, in rupees, that she had spent in buying almonds is *(CAT 2022, Slot 1)*

A. 1440 B. 1176 C. 1680 D. 2520

98. The monthly sales of a product from January to April were 120, 135, 150 and 165 units, respectively. The cost price of the product was Rs. 240 per unit, and a fixed marked price was used for the product in all the four months. Discounts of 20%, 10% and 5% were given on the marked price per unit in January, February and March, respectively, while no discounts were given in April. If the total profit from January to April was Rs. 138825, then the marked price per unit, in rupees, was *(CAT 2025, Slot 3)*

A. 525 B. 520 C. 510 D. 515

Ratio, Proportion & Variation (4 questions)

99. A stall sells popcorn and chips in packets of three sizes: large, super, and jumbo. The numbers of large, super, and jumbo packets in its stock are in the ratio 7 : 17 : 16 for popcorn and 6 : 15 : 14 for chips. If the total number of popcorn packets in its stock is the same as that of chips packets, then the numbers of jumbo popcorn packets and jumbo chips packets are in the ratio *(CAT 2017, Slot 1)*

A. 1 : 1 B. 8 : 7 C. 4 : 3 D. 6 : 5

100. Amala, Bina, and Gouri invest money in the ratio 3 : 4 : 5 in fixed deposits having respective annual interest rates in the ratio 6 : 5 : 4. What is their total interest income (in Rs) after a year, if Bina's interest income exceeds Amala's by Rs 250? *(CAT 2019, Slot 1)*

A. 7000 B. 6000 C. 6350 D. 7250

101. In an examination, Rama's score was one-twelfth of the sum of the scores of Mohan and Anjali. After a review, the score of each of them increased by 6. The revised scores of Anjali, Mohan, and Rama were in the ratio 11 : 10 : 3. Then Anjali's score exceeded Rama's score by *(CAT 2019, Slot 2)*

A. 26 B. 32 C. 24 D. 35

102. Pinky is standing in a queue at a ticket counter. Suppose the ratio of the number of persons standing ahead of Pinky to the number of persons standing behind her in the queue is 3 : 5. If the total number of persons in the queue is less than 300, then the maximum possible number of persons standing ahead of Pinky is *(CAT 2022, Slot 1)*

Averages (21 questions)

103. An elevator has a weight limit of 630 kg. It is carrying a group of people of whom the heaviest weighs 57 kg and the lightest weighs 53 kg. What is the maximum possible number of people in the group? *(CAT 2017, Slot 1)*

104. The average height of 22 toddlers increases by 2 inches when two of them leave this group. If the average height of these two toddlers is one-third the average height of the original 22, then the average height, in inches, of the remaining 20 toddlers is *(CAT 2017, Slot 2)*

A. 30 B. 28 C. 32 D. 26

105. In an apartment complex, the number of people aged 51 years and above is 30 and there are at most 39 people whose ages are below 51 years. The average age of all the people in the apartment complex is 38 years. What is the largest possible average age, in years, of the people whose ages are below 51 years? *(CAT 2018, Slot 1)*

A. 25 B. 26 C. 27 D. 28

106. The arithmetic mean of x, y and z is 80, and that of x, y, z, u and v is 75, where u = (x + y)/2 and v = (y + z)/2. If $x \geq z$, then the minimum possible value of x is *(CAT 2018, Slot 2)*

107. Let A, B and C be three positive integers such that the sum of A and the mean of B and C is 5. In addition, the sum of B and the mean of A and C is 7. Then the sum of A and B is *(CAT 2020, Slot 1)*

A. 6 B. 4 C. 7 D. 5

108. In a group of 10 students, the mean of the lowest 9 scores is 42 while the mean of the highest 9 scores is 47. For the entire group of 10 students, the maximum possible mean exceeds the minimum possible mean by *(CAT 2020, Slot 2)*

A. 5 B. 3 C. 4 D. 6

109. A batsman played n + 2 innings and got out on all occasions. His average score in these n + 2 innings was 29 runs and he scored 38 and 15 runs in the last two innings. The batsman scored less than 38 runs in each of the first n innings. In these n innings, his average score was 30 runs and lowest score was x runs. The smallest possible value of x is *(CAT 2020, Slot 3)*

A. 1 B. 3 C. 2 D. 4

110. Suppose hospital A admitted 21 less Covid infected patients than hospital B, and all eventually recovered. The sum of recovery days for patients in hospitals A and B were 200 and 152, respectively. If the average recovery days for patients admitted in hospital A was 3 more than the average in hospital B then the number admitted in hospital A was *(CAT 2021, Slot 1)*

111. Onion is sold for 5 consecutive months at the rate of Rs 10, 20, 25, 25, and 50 per kg, respectively. A family spends a fixed amount of money on onion for each of the first three months, and then spends half that amount on onion for each of the next two months. The average expense for onion, in rupees per kg, for the family over these 5 months is closest to *(CAT 2021, Slot 1)*

A. 18 B. 26 C. 16 D. 20

112. In a football tournament, a player has played a certain number of matches and 10 more matches are to be played. If he scores a total of one goal over the next 10 matches, his overall average will be 0.15 goals per match. On the other hand, if he scores a total of two goals over the next 10 matches, his overall average will be 0.2 goals per match. The number of matches he has played is *(CAT 2021, Slot 2)*

113. The arithmetic mean of scores of 25 students in an examination is 50. Five of these students top the examination with the same score. If the scores of the other students are distinct integers with the lowest being 30, then the maximum possible score of the toppers is *(CAT 2021, Slot 3)*

114. The average of three integers is 13. When a natural number n is included, the average of these four integers remains an odd integer. The minimum possible value of n is *(CAT 2022, Slot 1)*

A. 3 B. 4 C. 5 D. 1

115. Five students, including Amit, appear for an examination in which possible marks are integers between 0 and 50, both inclusive. The average marks for all the students is 38 and exactly three students got more than 32. If no two students got the same marks and Amit got the least marks among the five students, then the difference between the highest and lowest possible marks of Amit is *(CAT 2022, Slot 2)*

A. 21 B. 24 C. 20 D. 22

116. Manu earns ₹4000 per month and wants to save an average of ₹550 per month in a year. In the first nine months, his monthly expense was ₹3500, and he foresees that, tenth month onward, his monthly expense will increase to ₹3700. In order to meet his yearly savings target, his monthly earnings, in rupees, from the tenth month onward should be *(CAT 2022, Slot 2)*

A. 4200 B. 4400 C. 4300 D. 4350

117. The average of a non-decreasing sequence of N numbers $a_1, a_2, \ldots, a_N$ is 300. If a_1 is replaced by $6a_1$, the new average becomes 400. Then, the number of possible values of a_1 is *(CAT 2022, Slot 2)*

118. Consider six distinct natural numbers such that the average of the two smallest numbers is 14, and the average of the two largest numbers is 28. Then, the maximum possible value of the average of these six numbers is *(CAT 2022, Slot 3)*

A. 22.5 B. 23.5 C. 24 D. 23

119. In an examination, the average marks of students in sections A and B are 32 and 60, respectively. The number of students in section A is 10 less than that in section B. If the average marks of all the students across both the sections combined is an integer, then the difference between the maximum and minimum possible number of students in section A is *(CAT 2022, Slot 3)*

120. In an examination, the average marks of 4 girls and 6 boys is 24. Each of the girls has the same marks while each of the boys has the same marks. If the marks of any girl is at most double the marks of any boy, but not less than the marks of any boy, then the number of possible distinct integer values of the total marks of 2 girls and 6 boys is *(CAT 2023, Slot 1)*

A. 19 B. 21 C. 20 D. 22

121. There are four numbers such that average of first two numbers is 1 more than the first number, average of first three numbers is 2 more than average of first two numbers, and average of first four numbers is 3 more than average of first three numbers. Then, the difference between the largest and the smallest numbers, is *(CAT 2024, Slot 1)*

122. The average of three distinct real numbers is 28. If the smallest number is increased by 7 and the largest number is reduced by 10, the order of the numbers remains unchanged, and the new arithmetic mean becomes 2 more than the middle number, while the difference between the largest and the smallest numbers becomes 64. Then, the largest number in the original set of three numbers is *(CAT 2024, Slot 3)*

123. In a school with 1500 students, each student chooses any one of the streams out of science, arts, and commerce, by paying a fee of Rs 1100, Rs 1000, and Rs 800, respectively. The total fee paid by all the students is Rs 15,50,000. If the number of science students is not more than the number of arts students, then the maximum possible number of science students in the school is *(CAT 2025, Slot 3)*

Answer key

Part A: Classwork (1 to 79)

1. A	**2.** B	**3.** D	**4.** 80	**5.** C	**6.** 20	**7.** C
8. A	**9.** C	**10.** C	**11.** C	**12.** C	**13.** B	**14.** D
15. 55	**16.** 340	**17.** A	**18.** D	**19.** B	**20.** D	**21.** B
22. B	**23.** 2160	**24.** A	**25.** D	**26.** 20000	**27.** B	**28.** C
29. D	**30.** 175	**31.** B	**32.** B	**33.** 1000	**34.** 407	**35.** D
36. 12	**37.** B	**38.** 20	**39.** 9	**40.** D	**41.** 4000	**42.** 16000
43. 90000	**44.** 20920	**45.** 9	**46.** A	**47.** A	**48.** 121000	**49.** A
50. B	**51.** D	**52.** 272	**53.** C	**54.** 800	**55.** A	**56.** B
57. D	**58.** D	**59.** 66	**60.** D	**61.** C	**62.** 43200	**63.** B
64. B	**65.** B	**66.** C	**67.** C	**68.** A	**69.** A	**70.** 18
71. 49	**72.** B	**73.** D	**74.** D	**75.** 60	**76.** B	**77.** A
78. C	**79.** A					

Part B: Practice (80 to 123)

80. 20	**81.** 70000	**82.** B	**83.** D	**84.** A	**85.** C	**86.** B
87. A	**88.** 250	**89.** D	**90.** 900	**91.** B	**92.** B	**93.** A
94. A	**95.** D	**96.** 160	**97.** C	**98.** A	**99.** A	**100.** D
101. B	**102.** 111	**103.** 11	**104.** C	**105.** D	**106.** 105	**107.** A
108. C	**109.** C	**110.** 35	**111.** A	**112.** 10	**113.** 92	**114.** C
115. C	**116.** B	**117.** 14	**118.** A	**119.** 63	**120.** B	**121.** 15
122. 70	**123.** 700					